NNN-D-1478B
March 7, 1978
SUPERSEDING
Fed. Spec. NNN-D-1478A
February 2, 1974

FEDERAL SPECIFICATION

DISH, CULTURE, PETRI (GLASS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers glass petri dishes for use in performing bacteriological culture growth and biological assays.

1.2 Classification. The dishes covered by this specification shall be of the following types, classes, and sizes as specified (see 6.2).

1.2.1 Types.

Type I - Blown.
Type II - Pressed.

1.2.2 Classes.

Class 1 - Complete dish, top section and bottom section.
Class 2 - Top section only.
Class 3 - Bottom section only.

1.2.3 Sizes.

60 x 15 mm.
100 x 10 mm.
100 x 15 mm.
100 x 20 mm.
100 x 22 mm.
150 x 15 mm.
150 x 20 mm.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 6640

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American Society for Testing and Materials (ASTM) Standards:

C 148 - Polariscopic Examination of Glass Containers.
C 162 - Terms Relating to Glass and Glass Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. All classes shall be of transparent borosilicate glass.

3.2 Construction. The dish (top and bottom section) shall be of conventional shape with sides approximately perpendicular to the circular surface. Type I petri dishes shall have reinforced beaded edges to provide increased resistance to mechanical breakage and effect centering of the bottom dish inside the cover. The type II bottom dish shall have a flat inside bottom surface. This bottom dish shall have a raised outer edge to provide a level three-point support. The outer side of this bottom shall have a sandblast spot 120 square millimeter (mm) in area.

3.3 Sizes and dimensions. Petri dishes covered by this specification shall be furnished in the sizes and dimensions shown in table I, as specified.

TABLE I. Sizes and dimensions

Nominal size mm.	Minimum wall thickness mm.	Top 1/ Inside Depth mm.	Top 1/ Inside Dia. mm.	Bottom 1/ Outside Depth mm.	Bottom 1/ Outside Dia. mm.
60 x 15	1	14	57.5	16	57
100 x 10	1	9	95.5	11	95
100 x 15	1	14	95.5	16	95
100 x 20	1	19	95.5	21	95
100 x 22 2/	1	--	----	22	95
150 x 15	1	14	145.5	16	145
150 x 20	1	19	145.5	21	145

1/ Tolerances: Depth, top - 2 mm, bottom + 2 mm. Diameter, top + 3 mm, bottom - 3 mm.
2/ Type II, class 3.

3.4 Surface. The surface of type I dishes shall have no degree of curvature, convex, or concave, exceeding 1.6 mm. The surface of type II dishes shall have no degree of curvature, convex, or concave exceeding 0.5 mm.

3.5 Uniformity. The dishes of any one size shall be fabricated so that the tops and bottoms are interchangeable.

3.6 Annealing. When examined with a polariscope as specified in 4.4.1, there shall be no ring strain greater than temper no. 1. Longitudinal strain, if present, shall be faint and highly diffused.

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3.7 Thermal shock test. The petri dishes shall not break nor shall they show evidence of crystallization or clouding upon removal from the autoclave or from the cold immersion bath, when tested as specified in 4.4.2.

3.8 Identification markings. Each petri dish shall be legibly and permanently marked with name or registered trademark of the manufacturer. Tops and bottoms shall be distinguished from each other (by a method determined by each manufacturer to facilitate error-free sorting and permit easy identification of parts when assembling pairs).

3.9 Unit of issue. One dish complete (top and bottom) constitutes one unit of issue.

3.10 Workmanship. The dishes shall be uniform as to size and thickness, and shall be free from such defects as bubbles, seeds, blisters, striae, or optical imperfections which would impair their efficiency or adversely affect their serviceability. The glass shall have the ability to withstand repeated sterilization, wet or dry, without loss of clarity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Unit of product for inspection purposes shall consist of the individual top or bottom of a petri dish and a complete dish, top and bottom.

TABLE II. Sampling criteria

Component (unit of product)	Characteristics	Rqmt.	Test procedure	Inspection level	AQL (percent defective)
End item	Thermal shock	3.7	4.4.2	S-3	4.0
	Strain test	3.6	4.4.1	S-2	1.0

4.3 Examination. The tops and bottoms of the petri dishes shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

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Categories and defects 1/	
Major	
101	Components missing (top and bottom where applicable).
102	Petri dish (top and bottom) not free of cracks, chips, or breaks.
103	Top or bottom of petri dishes not free of crizzels or checks.
104	Tops and bottoms of petri dishes not interchangeable.
105	Petri dishes not clean, dry, and ready for use.
106	Inside bottom surface of type II bottom dish not flat.
107	Type II bottom dish not fabricated of heavy walls.
108	Outer edges of type II bottom not raised to provide a level three point support.
109	Top or bottom of petri dishes not free of sharp edges or protrusions.
110	Reinforcing beaded edges on top or bottom dishes not present (type I).
111	Tops or bottoms not free of scratches, bubbles, and stones.
112	Tops or bottoms not free of blisters, striae, and seeds.
113	Tops and bottoms not free of marred areas or optical imperfections.
Minor	
201	Dishes not uniform in thickness.
202	Identification marking not present, complete, correct, legible, or permanent.

1/ Examination shall not be restricted to the defects listed above. Definitions of glass defects shall be in accordance with ASTM C 162.

4.3.2 Dimensional examination. The petri dishes shall be examined for defects in dimensions. Any dimensions not within tolerance specified herein shall be classified as a defect.

4.4 Tests.

4.4.1 Annealing. The petri dishes shall be tested in accordance with method A or B of ASTM C 148.

4.4.2 Thermal shock test. The petri dishes shall be heated in an autoclave at 15 pounds pressure for 15 minutes. They shall be removed from the autoclave and quickly plunged into water at a temperature of 3° to 7° Centigrade (C).

4.5 Examination of preparation for delivery requirements. An examination shall be made to determine compliance with the packaging, packing, and marking requirements of section 5. Defects shall be scored in accordance with table III. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for examination at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 expressed in terms of percent defective.

TABLE III. Examination of preparation for delivery

Examine	Defects
Material	Any nonconforming component, component missing, damaged, or otherwise defective.
Container	Not as specified.
Content	Not as specified.
Workmanship	Inadequate application of components container bulged or distorted.
Marking	Omitted, incorrect; illegible; improper size, location, sequence of method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit container.

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5.1.1.1.1 Class 1. The culture dish of one type only shall have the top and bottom sections wrapped separately in tissue paper or other suitable material and then assembled into a complete dish.

5.1.1.1.2 Classes 2 and 3. The class 2 or class 3 section shall be wrapped in tissue paper or other suitable material.

5.1.1.2 Intermediate container. Unless otherwise specified, twelve unit containers of one description only shall be packaged in a corrugated fiberboard box. The box shall be secured to prevent accidental opening.

5.1.2 Commercial. Culture dishes of one type and class only and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling, and storage.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified, twelve intermediate containers shall be packed in a weather resistant corrugated fiberboard box. The box shall be fitted with partitions or dividers to protect the item. The box shall be closed with adhesive, staples or tape to secure the flaps.

5.2.2 Commercial. The packaged items shall be packed in a fiberboard box to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization. When shipments to Government depots are full car or truckload, the shipping containers shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. The unit, intermediate, and shipping containers and unitized loads, as applicable, shall be marked as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. Petri dishes are intended for use in laboratories for bacterial culture growth and biological assays. Pressed glass dishes offer greater resistance to mechanical shock and are of heavier construction than are the blown dishes. Nominal size 100 x 22, type II, bottom, pressed, (improved), accommodates nominal size 100 x 15 or 100 x 20 top, blown.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and size as applicable (see 1.2).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) Quantities required (see 5.1.1.2 and 5.2.1).
(e) Marking required (see 5.4).

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Review activities:

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Army - MD
Navy - MS
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DLA - DM

Preparing Activity:

GSA - FSS

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